

Technology Validation and Start Up Fund

Round 42 Proposal Evaluations

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Submitted to:
Lydia L. Mihalik
Director, Ohio Development of Development
Chair, Ohio Third Frontier Commission



TECHNOLOGY VALIDATION AND STARTUP FUND

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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 12 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the five-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (www.Redwdinnnov.com). Details of the TVSF program and the review process are provided in Appendix 2.

Twenty-one (21) TVSF Round 42 applications, totaling \$4,328,300, were received and evaluated: two Phase 1 applications totaling \$538,300 (2 Phase 1, Option 1 applicants) and 19 Phase 2 applications totaling \$3,790,000.

Funding is recommended for one Phase 1, Option 1 applications totaling \$200,000 and nine Phase 2 applications for a total of \$1,800,000 yielding a combined total of \$2,000,000. This translates to a 48% recommended application funding rate for Phase 1/Phase 2 TVSF Round 42, compared to the average of 51% over all 42 TVSF rounds. The recommended approval rate for Phase 2 proposals in this round is 47%.

2) Evaluation Results

Summaries of the evaluations of the proposals for Round 42, incorporating the interviews, and funding recommendations are shown below in Tables 1 and 2. Round 42 interviews were conducted via video conference. The total recommended funding for Phase 1 and Phase 2 projects is \$200,000, and \$1,800,000, respectively. Note that the Table 1 and 2 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Selection Committee which is weighted at 20 is four times as wide as Project Selection which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 1 and Phase 2 proposal may be found in Section 3 of this report.

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Table 1
Phase 1 Proposal Evaluation and Funding Recommendation
TVSF Round 42

Proposal #	Applicant	Requested Funding	Recommended Funding	Selection Committee	Deal Flow & Budget Strategy	External Participation	Track Record	Metrics	Project Management & Experience	Project Selection Process
FY26-5198	Research Institute at Nationwide Childrens Hospital	\$ 200,000	\$ 200,000							
	Sub-Total	\$ 200,000	\$ 200,000							
FY26-5197	Miami (OH) University	\$ 338,300	\$ -							
	Sub-Total	\$ 338,300	\$ -							
	Total	\$ 538,300	\$ 200,000	Column width is proportional to score weighting in each category						
Evaluation Scale				Weak	Meets	Exceeds				

Option 1: \$200,000 to \$500,000, 1:1 cost share, 1 year to allocate funds, 2 year project period.
Option 2: \$1,000,000, 1:1 cost share, 2 years to allocate funds, 3 year project period.
Option 3: \$100,000, internship component, 1 year to allocate funds, 2 year project period.

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Please note: The applicants are listed in the following table by recommended and not recommended for funding. In each section, the applicants are listed in numeric order by proposal number.

Table 2
Phase 2 Proposal Evaluation and Funding Recommendation
TVSF Round 42

Proposal Number	Lead Applicant	Requested Funding	Recommended Funding	Team	Opportunity/Market Size	IP Protection	Proof of Concept	Potential Investor/ Business Partner Engagement	Business Model	Project Plan/Budget Narrative	Growth Plan in Ohio	ESP Interaction
FY26-5171	ASAKE Biotechnology, LLC	\$ 200,000	\$ 200,000									
FY26-5173	Carbon Copy Assets Inc.	\$ 200,000	\$ 200,000									
FY26-5175	CYRANNUS INC	\$ 200,000	\$ 200,000									
FY26-5178	LegacyAcres, LLC	\$ 200,000	\$ 200,000									
FY26-5181	necoTECH, LLC	\$ 200,000	\$ 200,000									
FY26-5184	Relantic, Inc.	\$ 200,000	\$ 200,000									
FY26-5185	Saturn Sports LLC	\$ 200,000	\$ 200,000									
FY26-5186	SenselCs Corporation	\$ 200,000	\$ 200,000									
FY26-5187	Shaun Francis LLC	\$ 200,000	\$ 200,000									
Sub-Total, \$		\$ 1,800,000	\$ 1,800,000									
FY26-5172	BG Biologics LLC	\$ 200,000	\$ -									
FY26-5174	Cardinex LLC	\$ 190,000	\$ -									
FY26-5176	Gimbal Innovations, Inc.	\$ 200,000	\$ -									
FY26-5177	iBlades.ai	\$ 200,000	\$ -									
FY26-5179	MAK Engineering Services	\$ 200,000	\$ -									
FY26-5180	MiniOrtho Inc.	\$ 200,000	\$ -									
FY26-5182	Quality Life Sciences Engineering LLC	\$ 200,000	\$ -									
FY26-5183	Red House Studio LLC	\$ 200,000	\$ -									
FY26-5188	SLIPLY	\$ 200,000	\$ -									
FY26-5189	Wokkah	\$ 200,000	\$ -									
Sub-Total, \$		\$ 1,990,000	\$ -	Column width is proportional to score weighting in each category								
Total, \$		\$ 3,790,000	\$ 1,800,000									
Evaluation Scale				Weak	Meets	Exceeds						

Table 3 lists the funding approval rate by TVSF round. This round's approval rate of 48% of the total submitted proposals is on the high side of the average over all rounds. The Phase 1 applications recommended for approval comprised 5% of the total submittals for this round. The historical range of recommendations for individual rounds has spanned 27 – 100%, with an average of 51%.

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Table 3. TVSF Approval Rate by Round

TVSF Round 42 Funding & Approval Rate by Round					
Round	\$ Recommended	% Recommended	Round	\$ Recommended	% Recommended
	Phase 1 & Phase 2			Phase 1 & Phase 2	
1 (APR 2012)	\$950,000	35%	20 (NOV 2019)	\$1,350,000	43%
2 (AUG 2012)	\$900,000	52%	21 (FEB 2020)	\$3,944,000	56%
3 (DEC 2012)	\$610,000	44%	22 (JUN 2020)	\$1,398,630	53%
4 (JUN 2013)	\$864,000	30%	23 (DEC 2020)	\$900,000	50%
5 (FEB 2014)	\$1,462,000	46%	24 (MAR 2021)	\$2,092,900	55%
6 (JUN 2014)	\$998,000	39%	25 (JUN 2021)	\$800,000	75%
7 (OCT 2014)	\$1,100,000	57%	26 (OCT 2021)	\$1,700,000	55%
8 (FEB 2015)	\$710,000	37%	27 (FEB 2022)	\$850,000	43%
9 (JUN 2015)	\$550,000	31%	28 (APR 2022)	\$2,499,976	64%
10 (DEC 2015)	\$925,000	38%	29 (JULY 2022)	\$850,000	100%
11 (APR 2016)	\$1,239,000	46%	30 (OCT 2022)	\$3,700,000	71%
12 (OCT 2016)	\$3,537,269	46%	31 (JAN 2023)	\$100,000	50%
13 (MAR 2017)	\$1,567,500	38%	32 (APR 2023)	\$850,000	64%
14 (SEP 2017)	\$498,832	27%	33 (JULY 2023)	\$1,100,000	73%
15 (DEC 2017)	\$2,250,000	38%	34 (OCT 2023)	\$250,000	33%
16 (MAR 2018)	\$2,098,600	52%	35 (JAN 2024)	\$800,000	59%
17 (SEP 2018)	\$2,100,000	42%	36 (APR 2024)	\$2,650,000	80%
18 (DEC 2018)	\$1,150,000	35%	37 (JULY 2024)	\$1,798,000	82%
19 (APR 2019)	\$2,250,000	43%	38 (OCT 2024)	\$6,475,000	65%
			39 (JAN 2025)	\$1,400,000	50%
			40 (APR 2025)	1,800,000	59%
			41 (JUL 2025)	1,162,956	40%
			42 (OCT 2025)	2,000,000	48%
			Total Funding	\$64,231,663	
			Average/Round	\$ 1,576,944	51%

Please note that the following abbreviations may be used in the one-page summaries listed below.
TAM- total addressable market; SAM- Serviceable Available Market; SOM- Serviceable Obtainable Market;
ERDC: Engineering Research and Development Center, U.S. Army Corp of Engineers; DEVCOM- U.S. Army
Combat Capabilities Development Command.

3) Proposal Summaries

Phase 1 Proposal Summary - Recommended for Funding

Proposal 26-5198	Research Institute of Nationwide Children's Hospital	Amount Requested: \$200,000
Prior Phase 1 Applications: Yes	<i>NCHRI_TV_SF Round 42</i>	Amount Recommended: \$200,000

TOTAL BUDGET: \$400,000

Institution Snapshot: The strategic vision of the program is to advance inventions/intellectual property along the commercialization pathway with the purpose of making early inventions both more attractive and less risky to potential licensing and/or business partners.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Selection Committee	The 5-member Committee has a well-rounded set of skills and experiences necessary to developing successful startup companies.
Y	Deal Flow; Budget Strategy	TVSF program is a subset of the Technology Development Fund. With 95 invention disclosures filed last year and 2 to 5 applications per TVSF funding cycle, the program has a very robust pipeline.
Y	External Participation	Application is modestly supportive of engaging outside entities in development work/tech validation. 3rd party use is not always an option, particularly if researcher is on the leading edge of her field.
G	Track Record	NCHRI has a strong track record of licensing, commercializing technologies, startups being acquired. Of the 19 TVSF funded projects 10 are still active, 3 closed, the rest licensed or in negotiation.
Y	Metrics	In answers to pre-interview questions, specific TVSF outcome metrics were identified.
Y	Project Management & Experience	Projects are actively monitored by staff of OTC. Assigned licensing associate is considered the project manager for project.
G	Project Selection	There is a clear 7-step process for applicants to be selected.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: The applicant has had a strong record of licensing and commercializing intellectual property. The pipeline is robust with 95 invention disclosures filed last year with 2 to 5 applications per TVSF cycle. In addition, follow on funding is available via the Catalyst Fund III at \$30M joint Rev1/OSU/NCH effort. This applicant is recommended for funding.

Proposal Summary - Phase 1 Not Recommended for Funding

Proposal 26-5197	Miami (OH) University	Amount Requested: \$338,300
Prior Phase 1 Applications: No	<i>Fisher Innovation College @Elm Tech Validation & Transfer Services</i>	Amount Recommended: \$0

TOTAL BUDGET: \$0

Institution Snapshot: Miami (OH) University's strategic vision is to accelerate the commercialization of high-potential technologies by supporting scalable ventures, facilitating industry-research matchmaking, and increasing the volume of validated IP entering the market.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Selection Committee	Committee members of the 8-member committee have backgrounds in finance, fund raising and general entrepreneurship. No deep, clear expertise and connections to the focus markets/technologies is apparent.
R	Deal Flow; Budget Strategy	Application cites expertise / ecosystem in six specific areas. It is not clear if/how all of the projects will have Miami generated IP. Other deal sources cited.
Y	External Participation	Independent 3rd-party oversight at key stages is mentioned. Examples are not provided.
Y	Track Record	This is a new application for Phase 1 funding. Piloted 15 projects over the last 18 months, initial results are encouraging.
R	Metrics	Some metrics mentioned. Did not list metrics that will be tracked with Phase 1 funding.
R	Project Management & Experience	Project management team is from the College@Elm team. Names / credentials are not provided.
Y	Project Selection	There is a 5-stage framework and scoring rubric to select projects.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendation: This is a new application from Miami University. Deal flow is cited through four pathways and may or may not utilize intellectual property from Miami University. The College@Elm office provides technology validation and transfer services to all four deal flow pathway participants. Answers to questions regarding the top 10 active research areas at Miami University listed intellectual property that is more than 20 years old and in the public domain. The applicant is not recommended for funding.

Phase 2 Proposal Summaries - Recommended for Funding

Proposal 26-5171	ASAKE Biotechnology, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Engineered Probiotic Supplement for Metabolic Health</i>
Biomedical/ Life Sciences		TechGROWTH Ohio

Company Snapshot: ASAKE Biotech proposes to develop a first-in-class, orally delivered probiotic supplement that supports weight management and healthy blood sugar levels by producing GLP-1-like activity in the gut.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Two OU faculty. CEO/scientist/former Director Edison Biotech Institute with MBA and entrepreneurial background. Second employee synthetic biology PhD. Both have regulatory and IP experience.
Y	Opportunity/Market Size	Probiotics market over \$62 billion. Metabolic wellness and weight management supplement over \$33 billion.
Y	Intellectual Property Protection	Provisional patent filing 9/12/25. In negotiations with OU for an Option for an Exclusive License.
Y	Proof of Concept	Currently TRL-4. TVSF funding would deliver MVP. Potentially ready for marketing and sales within 12-months.
Y	Potential Investor/ Business Partner Engagement	Has had conversations with VCs. Has identified potential manufacturing and distribution partners in Ohio.
Y	Business Model	Anticipate FDA GRAS acceptance. Plan to partner with Ohio firms for manufacturing and distribution and/or sell direct-to-consumer.
Y	Project Plan/ Budget Narrative	Budget and plan are appropriate for stage of development.
G	Growth Plan in Ohio	Committed to remaining in Ohio and using Ohio-based resources.
G	ESP Interaction	Working with TechGROWTH Ohio and OU Innovation Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: CEO has extensive experience in early-stage commercialization, including IP and regulatory. He was recruited to lead OU Edison Biotech Institute because of his background and experience in maturing biotechnology from academic sources. Product is similar to another GMO probiotic sold under FDA GRAS rules. If successful, ASAKE plans to use established Ohio probiotic manufacturing and distribution partners. Large market opportunity with platform potential.

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Proposal 26-5173	Carbon Copy Assets Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Cincinnati	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	TapIn
Sensors	Entrepreneurs' Center	

Company Snapshot: Carbon Copy Assets is developing a seamless, secure interactive virtual kiosk platform for restaurants and retail stores.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Team has strong experience in block chain and digital transactions. They are partnering to quickly learn and penetrate the quick service restaurant market.
G	Opportunity/Market Size	The initial target is the 2/3 of the US quick service restaurant market that does not have ordering kiosks. Market opportunity is in excess of \$2B.
Y	Intellectual Property Protection	A patent has been filed by University of Cincinnati for a secure, traceable transaction process based on block chain technology.
Y	Proof of Concept	Initial demonstrations have occurred. The product is at a TRL 7.
Y	Potential Investor/ Business Partner Engagement	Conversations are well underway to raise a \$2M round by early 2026.
Y	Business Model	A SaaS revenue model is planned for the initial market vertical of quick service restaurants.
Y	Project Plan/ Budget Narrative	Project plan is appropriate in focus and budget.
Y	Growth Plan in Ohio	Several current team members are out of State. All growth in Ohio.
Y	ESP Interaction	Carbon Copy has been engaged with the EC since June 2024.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: This application is recommended for funding. The experienced and skilled leadership team has done a fine job identifying an unmet need and defining a differentiated approach to enter the quick service restaurant market and scale from there.

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Proposal 26-5175	CYRANNUS INC	Amount Requested: \$200,000
<i>Licensing Institution</i>	U.S. Navy (NAVSEA/NSWC)	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Methods and Apparatus for Data Preprocessing</i>
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: Cyrannus is developing a single investment platform to help early-stage startups and early investors sell shares before a startup exits through an IPO or acquisition.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Members of the leadership team of Cyrannus have strong backgrounds in finance, VC and IT / AI.
Y	Opportunity/Market Size	The market opportunity is more than \$400M as there are many early-stage investors who seek simplified diligence and earlier liquidity.
Y	Intellectual Property Protection	The IP of interest is for preprocessing of unstructured data.
Y	Proof of Concept	The initial platform, PrimIQ, has been piloted for the last two years with promising results.
G	Potential Investor/ Business Partner Engagement	Founders are well networked; have \$100M in committed funds for investment in startups. Now raising \$1M to close this year.
G	Business Model	The business model is well considered with Angels and family offices for PrimIQ and broker/dealers for Secondary (SecuralQ) offering.
Y	Project Plan/ Budget Narrative	A creditable plan is given to move SecuralQ to commercial status.
Y	Growth Plan in Ohio	Three of the four principals reside in Ohio. All growth to be in Ohio.
Y	ESP Interaction	There is active engagement with the Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The strong Cyrannus team has developed a platform for early-stage investments that has the potential to greatly increase the efficiency of diligence and provide earlier liquidity to investors. Their focus is on the mid-west and this is a nice example of developing financial infrastructure for our state and region.

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Proposal 26-5178	LegacyAcres, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 41	<i>FARMS: Cultivating the Future of Farm Succession Planning</i>
Software/ Information Technology		Rev1

Company Snapshot: LegacyAcres is a digital platform company designing solutions to simplify farm succession planning by centralizing asset management, ownership tracking, and transition scenario planning.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The 2-member management team and 2 advisors have project management, agricultural financial and operational knowledge, fundraising expertise. Legacy Acres is relying on a software development team to develop the required product.
Y	Opportunity/Market Size	Total addressable market, service available market and service obtainable markets are defined and quantified.
Y	Intellectual Property Protection	Software algorithms developed at Ohio State University are licensed exclusively to Legacy Acres. Copyright will be filed upon receipt of fully developed product.
G	Proof of Concept	Legacy Acres has used FARMS spreadsheets with >100 people to help in succession planning. The market has a distinct need for technology that can help in succession planning.
Y	Potential Investor/ Business Partner Engagement	Pro forma indicates investor monies may not be required. Contingency planning identified three potential investor groups.
Y	Business Model	Growth is aggressive with product launch in Dec 2026 and selling 370 subscriptions in 2026. Pre orders to be taken after demos.
Y	Project Plan/ Budget Narrative	A budget and plan proposed to get to an MVP within 12 months.
G	Growth Plan in Ohio	Plans to hire 10 FTEs and partner with farm groups, universities.
Y	ESP Interaction	Interacting with local ESP.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: LegacyAcres has made significant progress in advancing the understanding of needs within the market, how to reach the market and has developed strong relationships with farm groups. The company is recommended for funding.

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Proposal 26-5181	necoTECH, LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	ERDC	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Activated Cold patch asphalt for superior Pavement Repair
Advanced Materials		Entrepreneurs' Center

Company Snapshot: necoTECH is developing COLDPOD, an activated cold patch material that delivers a rapid, durable, and user-friendly asphalt repair solution using proprietary activation chemistry.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Team has relevant market, business, technical, and fundraising experience for technology development. Recently secured over \$100M of private investment for production scale up of asphalt innovation products. VP Sales and VP Technology planned for 2026.
G	Opportunity/Market Size	Fragmented commercial market for repair of road surfaces in cold weather. 10% of Columbus market is ~\$4M. Improved estimates for US, Ohio SAM and SOM would help focus in medium term.
Y	Intellectual Property Protection	Negotiated partially exclusive license for ERDC patent application # 18/664225, signed agreement expected by end 2025.
Y	Proof of Concept	US Navy driven development, PoC for US Navy defined KPMs in field testing in 2023. Additional metrics developed to demonstrate production and environmental operating limits.
G	Potential Investor/ Business Partner Engagement	Discussions with investor are underway. Due diligence in process with additional investment companies.
Y	Business Model	Existing DoD contacts who have supported current development will be initial target before diversifying into commercial applications.
Y	Project Plan/ Budget Narrative	Activity will focus on addressing aspects identified in user field trials.
Y	Growth Plan in Ohio	HQ in Ohio, will manufacture in partners Ohio asphalt plants.
G	ESP Interaction	EC closely involved, future assistance activity planned.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: necoTECH is developing an activated cold patch material that delivers a rapid, durable and user-friendly repair solution using activation chemistry licensed from ERDC. US Navy trials have demonstrated practical viability and identified improvements for future production and operational performance limits. The management team has a proven history of commercialization and fund raising. Initial market focus will be with identified DoD applications; the medium-term success of expansion into future commercial applications would benefit from improved identification of market opportunities that can be obtained using established relationships with recognized local and national suppliers.

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Proposal 26-5184	Relantic, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Relantic
Software/ Information Technology		Rev1

Company Snapshot: The Relantic platform uses Artificial Intelligence to semantically weave together multiple modalities of data, accelerating workflows such as field reporting and claims processing to enable experts to transform into strategic decision makers.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	Relantic has a strong team of successful serial entrepreneurs and the needed AI / IT skills. The COO was a TVSF recipient in 2022 whose company had a successful exit in April 2025.
G	Opportunity/Market Size	The initial target market, Utility Company outage response, is in the billions of dollars. Other credible verticals are identified.
G	Intellectual Property Protection	Licensing discussions are underway on three issued patents owned by OSU and authored by one by of Relantic's founders. Detection of patent infringement is well considered.
Y	Proof of Concept	TVSF funding will advance TRL to a 5. This, combined with the team's credentials, will create value.
G	Potential Investor/ Business Partner Engagement	Pilots have been secured with strategic partners. Founders are also well networked with the funding community.
Y	Business Model	A well-accepted SaaS business model is planned.
Y	Project Plan/ Budget Narrative	Project plan is appropriate and prospective vendors are identified.
Y	Growth Plan in Ohio	Clear commitment to Ohio is given. Registered in Delaware.
Y	ESP Interaction	Engaged with Rev1 since July 2025.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: This application is enthusiastically recommended for funding. A strong team of serial entrepreneurs has developed an attractive AI based offering for a large initial market and has identified significant follow-on opportunities.

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Proposal 26-5185	Saturn Sports LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	US Army DEVCOM Chemical Biological Center	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 40	Chin Strap Monitoring System
Sensors	Entrepreneurs' Center	

Company Snapshot: Saturn Sports LLC is commercializing a football helmet chin strap to enhance player safety with visual and audible signals that indicate whether a chin strap tightly secures the helmet to the player.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Management team have demonstrated a firm commitment to the new company and commercialization of proposed IP. Advisory team has been enlisted to provide product development, engineering, business management and fund-raising expertise. Strategic hires planned.
Y	Opportunity/Market Size	SAM chin strap market is \$563M. SOM for American football chin strap market is \$165M. Adapting miniature electronic monitoring components to enhance helmet safety has social value.
Y	Intellectual Property Protection	In discussions with TechLink for licensing of US Patent 11,723,441. This is US Army DEVCOM IP. First adaptor can potentially secure a level of sustained competitive advantage.
G	Proof of Concept	Chin strap prototype has been designed, built and tested in lab bench scale demonstrating reliability and manufacturability - and providing a solid bill of materials and unit factory cost at scale.
G	Potential Investor/ Business Partner Engagement	Awarded \$76,000 in 2025 Flyer Pitch Competition plus SAFE round funding of \$100,000. Have had initial talks with seed fund entities.
G	Business Model	Sales channels include direct sales, e-commerce, and wholesale with tiered pricing. Solid, ongoing interactions with potential buyers.
G	Project Plan/ Budget Narrative	Design/initial production tasks planned with Converge Technologies.
Y	Growth Plan in Ohio	Project annual revenue of \$17M and 18 FT employees (Year 5).
G	ESP Interaction	Engaged with Entrepreneurs' Center.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Secured outside advisors with expertise in business management, engineering, product development, and Start-up fund raising. Commitment of the core management team and advisors has been demonstrated. Prototype has been successfully tested in lab environment. Solid interaction with potential customers. Funding support with University of Dayton Flyer Competition and SAFE round. Business model is well structured. Project plan is complete and logical. Cash projection is ambitious but reasonable with recognition of necessary equity investment to enable business growth.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5186	SenselCs Corporation	Amount Requested: \$200,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	AltiZoom for Altimetry
Sensors	Rev1	

Company Snapshot: SenselCs Corporation seeks to commercialize a next-generation chip for light detection and ranging (LiDAR) systems, with a primary focus on ranging and altimetry to measure altitude, elevation, and distance for terrain elevation modeling, inland water monitoring, floodplain mapping, and coastal elevation profiling.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The team has the relevant technical background and a proven history of obtaining non-dilutive federal funding. Fund raising and experience with direct product sales would be beneficial to support successful future growth.
Y	Opportunity/Market Size	Market for improved resolution in ranging and altimetry: geospatial, agricultural and environmental markets. Estimated market size for these areas is \$200-300M. Secondary target market automotive.
Y	Intellectual Property Protection	Exclusive worldwide license for U.S. Patent Application PCT/US2024/042252, filed August 14, 2024 by OSU.
Y	Proof of Concept	Improved resolution in ranging and altimetry is required for future use. Technology has demonstrated TRL 6 in testing on NASA contract, integration of licensed CFD IP will provide TRL 7.
Y	Potential Investor/ Business Partner Engagement	~\$2M required in Y4; plan to use grant funding, internal investment and federal programs. External investment considered.
Y	Business Model	Growth targets need < 2% of the automotive and altimetry market. No specific customers identified, market focus needs to be maintained.
Y	Project Plan/ Budget Narrative	Plan is aggressive but consistent with MVP in 12 months.
Y	Growth Plan in Ohio	\$12M revenues and 15 new FTEs by Y5 (Sales and Technical).
G	ESP Interaction	Engaged with Rev1 for review and future actions.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The AltiZoom chip will provide improved resolution for future altimeter and range measurement in geospatial, agricultural and environmental markets. Integration of the licensed technology will use considerable external goods and services; care will be needed to minimize unplanned delays. Detailed assessment of the initial target altimeter and future automotive markets would help identify specific target customers. Investment options for commercialization and future funds for production would benefit from fundraising experience. Successful future large volume direct sales will benefit from the early inclusion of specific expertise in product sales/marketing.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5187	Shaun Francis LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Research Institute at Nationwide Children's Hospital	Amount Recommended: \$200,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	ACNU Dermatology Products
Software/Information Technology		Rev1

Company Snapshot: The company plans to bring an acne fighting prescription-strength tretinoin to the market without a prescription by using a new FDA regulatory pathway called ACNU (Additional Conditions for Nonprescription Use). Completion of ACNU Tool could provide a platform for sale of other prescription drugs.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Founder is a serial entrepreneur with successful exits in other pharma-related and locally financed startups. He has been tracking ACNU law closely. Small cadre of highly qualified advisors/vendors.
Y	Opportunity/Market Size	Existing market is \$60M. ACNU access is expected to grow market significantly. ACNU Tool provides a platform for other drugs.
Y	Intellectual Property Protection	Trade Secret and/or copyright for ACNU software based on knowhow to be licensed from NCH.
Y	Proof of Concept	ACNU Tool is MVP. Human factors testing and FDA application to follow. Traditional drug clinical trials are not required.
G	Potential Investor/ Business Partner Engagement	Engaged with Rev1, institutional investors and career contacts.
Y	Business Model	Proforma reasonable and well developed. First mover and DTC models, appropriate for target market and online business.
Y	Project Plan/ Budget Narrative	Tasks and milestones, achievable within budget and timeframe.
Y	Growth Plan in Ohio	Ohio will serve as strategic hub for all operations.
G	ESP Interaction	Rev1 very supportive. Long-term engagement on strategy.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Founder has track record of successful launches of other pharma-related startups in Central Ohio. He has been closely monitoring the ACNU legislation as it developed. This new FDA pathway just passed into law, providing the first mover opportunity for which he and his investors have been waiting. The ACNU online tool will provide a portal for patients to purchase certain prescription drugs without a prescription, first in dermatology and later for other medical conditions. This application is recommended for funding.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal Summaries - Phase 2 Not Recommended for Funding

Proposal 26-5172	BG Biologics LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Bowling Green State University	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Control of Pythium Pathogens in greenhouse operations</i>
Biomedical/ Life Sciences		No ESP

Company Snapshot: BG Biologics proposes to develop a commercial formulation of PythiumX that is effective in mitigating pythium infections of plant roots in commercial hydroponic greenhouses.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	This is a one-person team with technical skills and background. Several advisors are listed however no information on how the advisors would contribute to the team is given. No business, fundraising, marketing or sales information listed.
R	Opportunity/Market Size	Hydroponics market identified as the market. The product is a treatment for fungal-like infection of plant roots. The TAM, SAM and SOM for this market sector are not identified or delineated.
Y	Intellectual Property Protection	IP is identified by patent application number. No discussion of how this patent would be used in the business.
Y	Proof of Concept	Applicant states that the product is at a TRL 7. A lab assay was developed and demonstrated effective dosing to 'colonize and kill every isolate (8 species)'.
R	Potential Investor/ Business Partner Engagement	Limited discussion of Potential Investor/Business Partner Engagement. Pro forma shows \$600K needed in 2026-7.
R	Business Model	Pro forma shows a net profit of over \$300K in 4 years and equity investment of \$600K.
R	Project Plan/ Budget Narrative	The timeframe listed in the table is over 3 years.
R	Growth Plan in Ohio	No growth plan is presented in the proposal.
R	ESP Interaction	No ESP listed. Some interaction with UT Business Incubator.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The BG Biologics team consists of one technically competent individual. The market opportunity needs to be correctly identified and quantified. The RFP provides suggestions for each section of the proposal to be submitted. Recommend adding necessary experience to management team and working closely with an ESP to guide the company in this application process and the future success of the company.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5174	Cardinex LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Toledo	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	AORTIC PUNCH
Biomedical/ Life Sciences		No ESP

Company Snapshot: Cardinex proposes to develop the One-Step Aortic Punch to simplify coronary artery bypass graft (CABG) surgery and improve patient outcomes.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Insufficient business and medical device development experience. Inventor/founder/professor lead was absent from interview. Recent BS grad tech lead. Vascular surgeon advisor.
R	Opportunity/Market Size	Cardiac surgery addressable market is small. Possible additional future markets small to moderate.
Y	Intellectual Property Protection	Finalizing an Option Agreement with the University of Toledo for rights to one patent application.
R	Proof of Concept	Currently TRL-2. Unlikely to achieve MVP ready for clinical testing in 12 months.
R	Potential Investor/ Business Partner Engagement	Commitment from Angel is insufficient. Amount budgeted for R&D is significantly underestimated. SBIR is not a viable pathway to fund.
R	Business Model	Incremental improvement to existing CABG workflow. Timelines and budgets are unrealistic. Plans are not well developed.
R	Project Plan/ Budget Narrative	Costs and schedules are underestimated.
Y	Growth Plan in Ohio	Committed to remaining in Ohio. Seeking Ohio vendors.
R	ESP Interaction	Limited help from University of Toledo Business Incubator.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Inexperienced business team. Principal absent from interview. Niche market opportunity is small. Timelines are unrealistically short. Funding estimates are insufficient to achieve goals. The team may benefit from additional proposal help from University of Toledo and JumpStart ESPs.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5176	Gimbal Innovations, Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Cincinnati	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 41	M10180 Prototype 2
Advanced Manufacturing	Cincy Tech	

Company Snapshot: Gimbal Innovations, Inc. is a Cincinnati-based startup company dedicated to building high-performance motion platforms which augment existing flight simulator VR technologies.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Two of the three members of the management team were on the interview. The CEO was not present.
G	Opportunity/Market Size	Five potential target markets are identified. Initial entry targets Home-Based Entertainment with a total available market estimated at more than \$1B and a Service Available Market of \$112M annually.
Y	Intellectual Property Protection	Base technology for the center of Gravity Management System was co-invented by Gimbal's founder and UC Aerospace Department. Additional patent applications have been filed by Gimbal.
G	Proof of Concept	The MVP for the HBE market has been validated through a testing and development program with over 1,000+ hours of operation and 7,000+ hours of load testing without major issues.
Y	Potential Investor/ Business Partner Engagement	Applicant states discussions have been held with a second angel investor and multiple VCs.
R	Business Model	Pro forma in application was 3 yr. Two additional requests made for clarification yielded 2 different pro formas from team.
R	Project Plan/ Budget Narrative	Key objectives-improve TRL 7 to 9 and MRL 6 to 10 in 6-12 months.
Y	Growth Plan in Ohio	Team is vested in Ohio and states that they plan to grow in Ohio.
Y	ESP Interaction	Gimbal has been interacting with the local ESP for over 2 years.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Gimbal plans to use the funds to produce 2 pilot units within 6 to 12 months and achieve MRL of 10. The answers to post interview questions indicated that material improvements need to be made to obtain a functioning prototype. The application and subsequent answers to pre and post interview questions did not present a coherent, compelling case for TVSF support.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5177	iBlades.ai	Amount Requested: \$200,000
<i>Licensing Institution</i>	National Security Administration	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Data Security for Smartphone Charging Connection
Software/ Information Technology		No ESP

Company Snapshot: iBlades is developing a smartphone case that protects mobile devices from data theft, hacking and spying in high-risk environments.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	The management team has the relevant technical skills and business experience with one of the team has a history of commercialization and fundraising.
R	Opportunity/Market Size	The global market for tamper-proof mobile security is identified as ~\$98B; no US market size is given. No estimate of SAM and SOM for a product with the proposed security is provided.
R	Intellectual Property Protection	A proposal has been submitted to the NSA for an exclusive license. The patent referenced is not an NSA patent.
Y	Proof of Concept	KPMs for critical customers are provided. The ManoTron product will integrate hardware at TRL 7 with licensed technology. The integration is currently at TRL 2-3
R	Potential Investor/ Business Partner Engagement	Additional investment of ~\$2M needed. No engagement with external investors noted.
R	Business Model	Revenue will be generated by hardware sales, recurring software licensing and services. No market justification for product pricing.
R	Project Plan/ Budget Narrative	Limited details provided. Integration will be by iBlades in Ogden, UT
R	Growth Plan in Ohio	1-2 direct jobs with revenues of ~\$17M after Y5.
R	ESP Interaction	No interaction with Ohio ESPs.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: iBlades.ai will provide secure mobile solution that protects communications and charging processes in high-risk environments. The global market for tamper-proof mobile security is identified as ~\$98B; focused market penetration would benefit from an assessment of the SAM and SOM for the US market. The IP position and ownership need to be reconciled prior to proceeding further. It is recommended that future discussions be held with the Ohio ESP network to help focus the proposal and identify quantifiable benefits that can accrue to the State of Ohio should future submissions be made.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5179	MAK Engineering Services	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Cincinnati	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	MAK Savonius Wind Turbine
Energy	No ESP	

Company Snapshot: MAK Engineering Services is focused on using improvements in the design of the Savonius wind turbine to increase energy production in medium/small scale renewable energy generation systems combining solar and wind.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Technical competence in appropriate areas. Business experience is suggested, but no quantifiable history of building successful commercial company or fundraising experience provided.
R	Opportunity/Market Size	Potential global and US market for wind power not given, no estimate of SAM or SOM for medium/small scale renewable energy sources provided.
Y	Intellectual Property Protection	Technology patented by University of Cincinnati: US patent #12/071,929 filed 7/13/23 issued. EU patent #22740116.3 filed 7/27/23, pending.
R	Proof of Concept	Claim to have "low energy" lab prototype, needs further development to scale up to commercial size unit.
R	Potential Investor/ Business Partner Engagement	No evidence of discussions with investors. Current plan relies on bootstrapping, loans and government grants.
R	Business Model	No details of pricing or profitability. Target markets are given by country, no unit size and manufacturing/delivery costs given.
R	Project Plan/ Budget Narrative	Insufficient details of proposed work. TVSF funds used internally.
R	Growth Plan in Ohio	40 FTEs, profits ~\$4M in Y5, manufacturing location not given.
R	ESP Interaction	No Ohio ESP. Worked with Ohio SBDC and APEX Accelerator.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: MAK Engineering Services wants to develop medium/small -scale renewable energy generation systems that combine solar and wind to provide discrete microgrids. The proposal would benefit from a clearer definition of the market size for the envisaged microgrids, associated production costs and sales price to allow the benefits to be clearly delineated from products currently available. The planned budget should be reconsidered to ensure that it meets TVSF guidelines for fund allocation between contractors and the applicant. Overall, the proposal would benefit from clear identification of how the product is differentiated from those of competitors and significant input from a relevant Ohio ESP.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5180	MiniOrtho Inc.	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Toledo	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 39	<i>Syndesmotic Ankle Fractures Fixation and Reconstruction System (SAFFRS)</i>
Biomedical/ Life Sciences		No ESP

Company Snapshot: MiniOrtho plans to commercialize a patented device, the syndesmotic ankle fracture fixation and reconstruction system (SAFFRS) designed to simplify the treatment of syndesmotic ankle fractures.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Technically strong academic team. Business leadership still lacking.
R	Opportunity/Market Size	Small to moderate market size for a medical device.
Y	Intellectual Property Protection	Licensor agrees to negotiate for one issued US patent if funding is obtained.
R	Proof of Concept	First working prototype in development. Successful completion of TVSF funding unlikely to yield MVP ready for clinical trials.
R	Potential Investor/ Business Partner Engagement	Conversations with potential investors or strategic partners were not substantive. Propose to fund with SBIR after TVSF.
R	Business Model	Propose to manufacture and sell the devices directly, or partner. Pro forma provided is incomplete. Near-term funding is absent.
Y	Project Plan/ Budget Narrative	Budget and plan are appropriate for stage of development.
Y	Growth Plan in Ohio	Registered in Ohio. Committed to staying in Ohio.
Y	ESP Interaction	Interaction with University of Toledo Business Incubator.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: MiniOrtho presents a new approach for certain ankle injuries. The technology is very immature, just building the first working prototype now. Business leadership is still lacking, especially medical device business development and fundraising. The business strategy is not well developed and lacks pro forma estimates. The founders should seek additional guidance from an ESP and others. Above all, MiniOrtho needs experienced business leadership. The technology may need to be advanced using other funds to provide sufficient progress to reach fundable level following TVSF.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5182	Quality Life Sciences Engineering LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	Bon Secours Mercy Health Foundation	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	ProDynaStrength Neck System
Biomedical/ Life Sciences		JumpStart

Company Snapshot: Quality Life Sciences Engineering LLC is developing a smart, data-driven neck strengthening system designed to reduce the risk of concussion by helping athletes safely build muscle strength in their necks.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	The "team" consists of the CEO and 1 person with PT experience in an advisory role. Limited experience in establishing and growing a successful business is provided. Proven skills in fundraising, sales, marketing and finance are missing.
R	Opportunity/Market Size	The global market for strength training equipment is identified. A target value for the US market has been estimated. Similar competitor products are available at a significantly lower cost
Y	Intellectual Property Protection	Patent application US 20230181097A1 has been filled for the United States, Europe, and Australia covering the head unit component of the product.
Y	Proof of Concept	Phase 1 prototype requires modification based on user input, development of critical KPMs and product redesign will be required for successful demonstration of MVP for the proposed application.
R	Potential Investor/ Business Partner Engagement	No discussions with investors have been initiated. Current plan is to explore nondilutive funding.
Y	Business Model	Sales will be through existing contacts. Hardware will be sold directly with analytical software provided for a recurring annual fee.
Y	Project Plan/ Budget Narrative	Identification of user needs will affect timely development of MVP.
R	Growth Plan in Ohio	Projected revenues of \$5M and 9 FTEs in Y5.
Y	ESP Interaction	Engaged with Jumpstart.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: In Phase 1 work, a smart, data-driven neck strengthening system identified the need to change market focus and product design, clarification of the impact on future development is required. Future success would benefit from including expertise in business operations and fundraising. Clarification of the market opportunity would be valuable and allow a comparison with existing products. The concept is at an early stage and requires clarification of the business model and clear delineation of a viable, competitive market focused product.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5183	Red House Studio LLC	Amount Requested: \$200,000
<i>Licensing Institution</i>	University of Akron Research Foundation	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	MILA: Melanin infused Lyophilized Aerogels
Advanced Materials		MAGNET

Company Snapshot: The goal of this Phase 2 project is to demonstrate that melanin-extraction aerogel materials derived from UA's patents can achieve—or exceed—the benchmark performance already shown by Redhouse materials, thereby unlocking substantial downstream licensing and manufacturing opportunities for UA.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	The team consists of the architect/founder of Redhouse Studio and UA professor who is co-inventor for aerogel technology. MAGNET and UA Research Foundation noted. No delineation of skills and team interactions given.
R	Opportunity/Market Size	Market Opportunity given as "radiation shielding markets exceed US \$14B". No TAM, SAM or SOM identified. No initial target market identified.
Y	Intellectual Property Protection	IP of UA patent and UA & NASA patent application identified by number. Redhouse stated as "owns proprietary IP on structural melanin composites".
R	Proof of Concept	No information given on Proof of Concept. No input from potential customers and key performance metrics given.
R	Potential Investor/ Business Partner Engagement	No section in proposal.
R	Business Model	No pro forma given. Proposal states "Business Model Revenue streams: melanin feedstock sales; aerogel panel licensing; . . ."
R	Project Plan/ Budget Narrative	Project plan in application form and proposal submitted do not align.
R	Growth Plan in Ohio	No growth plan, state a JV will be set up.
R	ESP Interaction	MAGNET contact listed. Not apparent if assisted with proposal.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: IP is available from UARF and discussions have been initiated with Redhouse. The application states the commercialization strategy of "license UA extraction IP to the joint startup". The initial target market is not identified or quantified. No pro forma is included. A "Suggested Business Revenue Table" was submitted. Should the applicant wish to resubmit, it would be extremely beneficial if the ESP was actively engaged with delineating the business and reviewing the proposal prior to submission.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5188	SLIPLY	Amount Requested: \$200,000
<i>Licensing Institution</i>	National Security Agency	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	<i>Method of Neighbor Embedding for OCR Enhancement</i>
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: Sliply is a mobile based SaaS platform for trucking fleets that uses proprietary OCR enhancement algorithms to digitalize handwritten job slips that are used for invoicing and to enable compliance with digital reporting standards.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	CEO founded and scaled software companies and oversaw software development/architecture teams. CFO has fund-raising and financial strategy expertise for software businesses. Lack of detail in proposal describing background and experience of the two co-founders.
R	Opportunity/Market Size	Ohio SAM \$50.4 MM (1,200 companies × \$42,000/year “opportunity cost”). Opportunity cost is calculated by the hours needed by driver to manual handle handwritten slips displacing job work @\$115/hour.
Y	Intellectual Property Protection	NSA US Patent 8,938,118 - Method of Neighbor Embedding for OCR Enhancement (2012). It is not clear if neighbor embedding method in patent has an advantage over current image enhancement methods.
R	Proof of Concept	Examples show enhancement of print not handwritten documents so accuracy uncertain for target application. MVP 2000+ job slips ~65% OCR accuracy. No data provided that IP will provide higher accuracy.
Y	Potential Investor/ Business Partner Engagement	CEO/CFO have established network of investors. Initial discussions with Enhanced Capital.
R	Business Model	License embedded versions to clients to manage subcontracts. Fee of 20% on total transactions. Revenue projection \$179 MM (5 year).
R	Project Plan/ Budget Narrative	BKL budget \$265,000. TVSF budget \$200,000. SOWs don't match.
Y	Growth Plan in Ohio	Projected 30 full-time staff in various roles based in Ohio.
Y	ESP Interaction	Proposal states EC Dayton have reviewed proposal.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Proposal and follow-up have not provided sufficient information on the expertise and background of management team. Black Kitty Lab tasks and price for contract funded work do not match in the proposal. Net profit of 20% “take rate” of total invoices is based on “foregone driving revenue” but is instead apparently a “finder’s fee” that is not fully explained in proposal. Patent has example of print, but not handwriting, enhancement to illustrate improved OCR accuracy. Proposal claims that licensed IP will increase handwriting OCR accuracy from ~ 65% to ~ 85% but does not provide any supporting data or show that current state-of art ORC handwriting software has been benchmarked.

TECHNOLOGY VALIDATION AND STARTUP FUND

Proposal 26-5189	Wokkah	Amount Requested: \$200,000
<i>Licensing Institution</i>	Naval Surface Warfare Center-Dahlgren Division	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Secure Product Launch Platform
Software/ Information Technology		Entrepreneurs' Center

Company Snapshot: Wokkah uses AI to generate boilerplate code from user-submitted requirements. It is integrating FLAME, a static analyzer, to detect and consolidate vulnerabilities in C/C++ and Java code.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Wokkah's enthusiastic founder was the only team member on the interview call. The founder has a great idea and a lot of development experience. Building an even closer relationship with a proven IT fundraiser / entrepreneur is recommended.
G	Opportunity/Market Size	The IT startup market is enormous and there are numerous inefficiencies targeted by Wokkah.
Y	Intellectual Property Protection	IP will consist of a licensed NSWC patent to harden code and a newly filed Wokkah patent.
Y	Proof of Concept	An initial partial pilot with six partners showed promising results with reductions in both cycle time and budget.
Y	Potential Investor/ Business Partner Engagement	Wokkah is in active discussions with two funds on annual raises of \$500K.
R	Business Model	Current business model articulated seems inconsistent – seeking both economies of scale and customization. COGS is very optimistic.
Y	Project Plan/ Budget Narrative	Project plan and budget are reasonable. Using UC Security Lab.
Y	Growth Plan in Ohio	One principal is located outside of Ohio. All new growth in Ohio.
Y	ESP Interaction	A long-term relationship exists with the EC in Dayton.
	Evaluator Recommendation	This application is not recommended for funding.

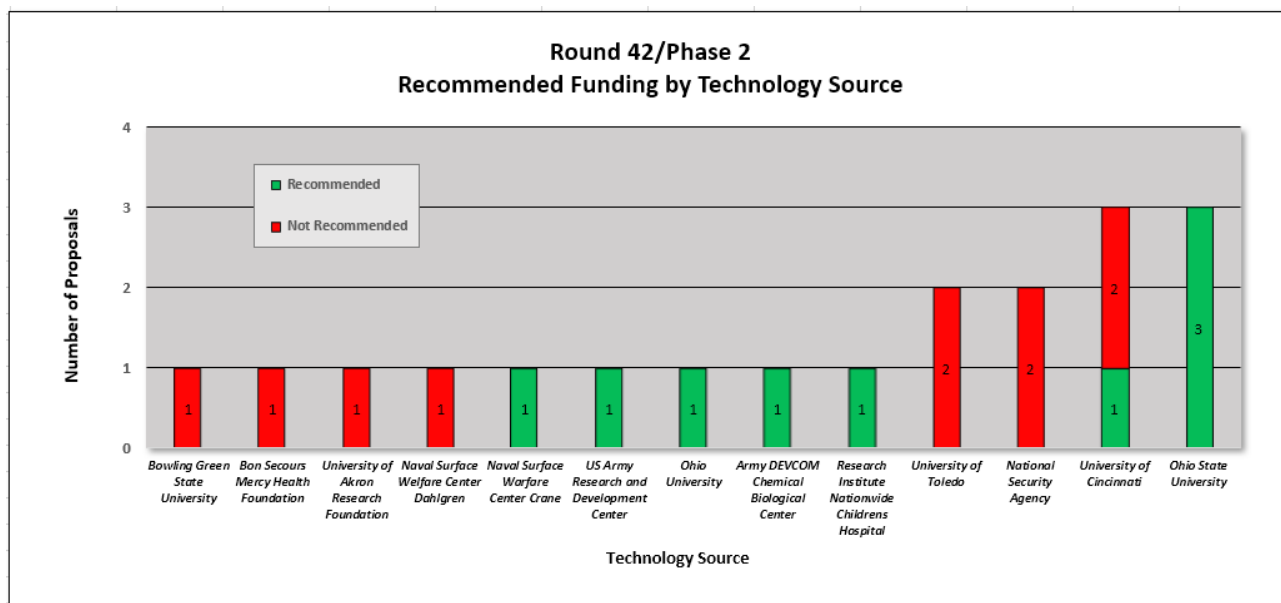
Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Wokkah has identified a significant opportunity and has a start on a strong product offering. If they reapply, they are encouraged to build out the relationship with their experienced entrepreneur and revisit the business model for consistency and reasonableness.

4) Round 42 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. There were 13 sources of technology this round, the most of the last 23 Rounds. Ohio State University and University of Cincinnati each provided three technologies; four of those six proposals are recommended. Two submissions each utilizing University of Toledo and National Security Agency technologies; none of the four proposals were recommended for funding. Single application submissions were received sourcing technology from Bowling Green State University, Bon Secours Mercy Health Foundation, University of Akron Research Foundation, Naval Surface Warfare Center Dahlgren, Naval Surface Warfare Center Crane, US Army Research and Development Center, Ohio University, US Army DEVCOM Chemical Biological Center and Nationwide Children's Hospital. One each from Naval Surface Warfare Center Dahlgren, Naval Surface Warfare Center Crane, US Army Research and Development Center, Ohio University, US Army DEVCOM Chemical Biological Center and Nationwide Children's Hospital are recommended for funding.

Figure 1. Round 42 Funding by Technology Source

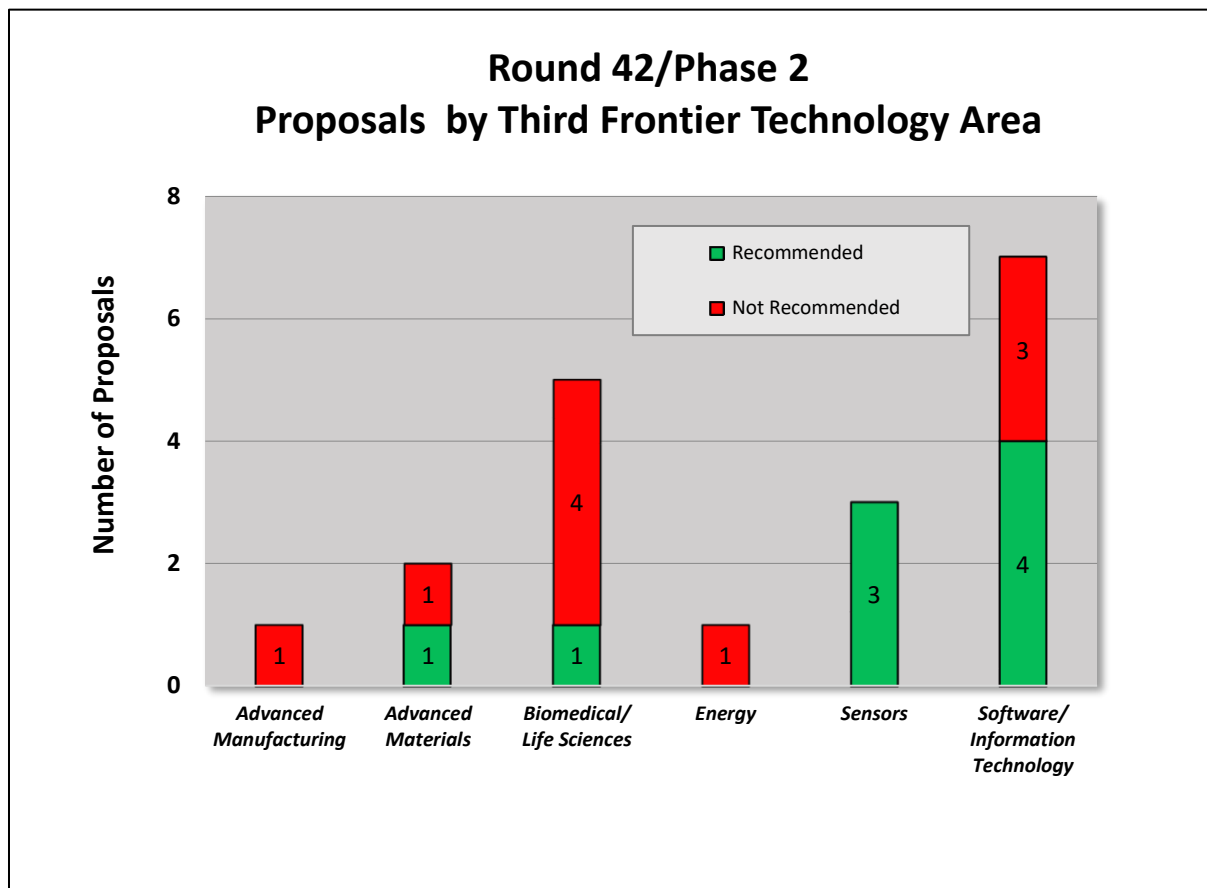


TECHNOLOGY VALIDATION AND STARTUP FUND

Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In this Round, seven of nineteen were in Software/Information Technology (37%), five of nineteen were in Biomedical/Life Sciences (26%), three of nineteen were in Sensors (16%), two of the nineteen (11%) were in Advanced Materials, and one of the nineteen was in Advanced Manufacturing and Energy (5%).

Four of the nineteen in Software/Information Technology (21%), three of the nineteen in Sensors (16%), and one of the nineteen in each of Advanced Manufacturing (5%) and Biomedical/Life Sciences (5%) were recommended for funding.

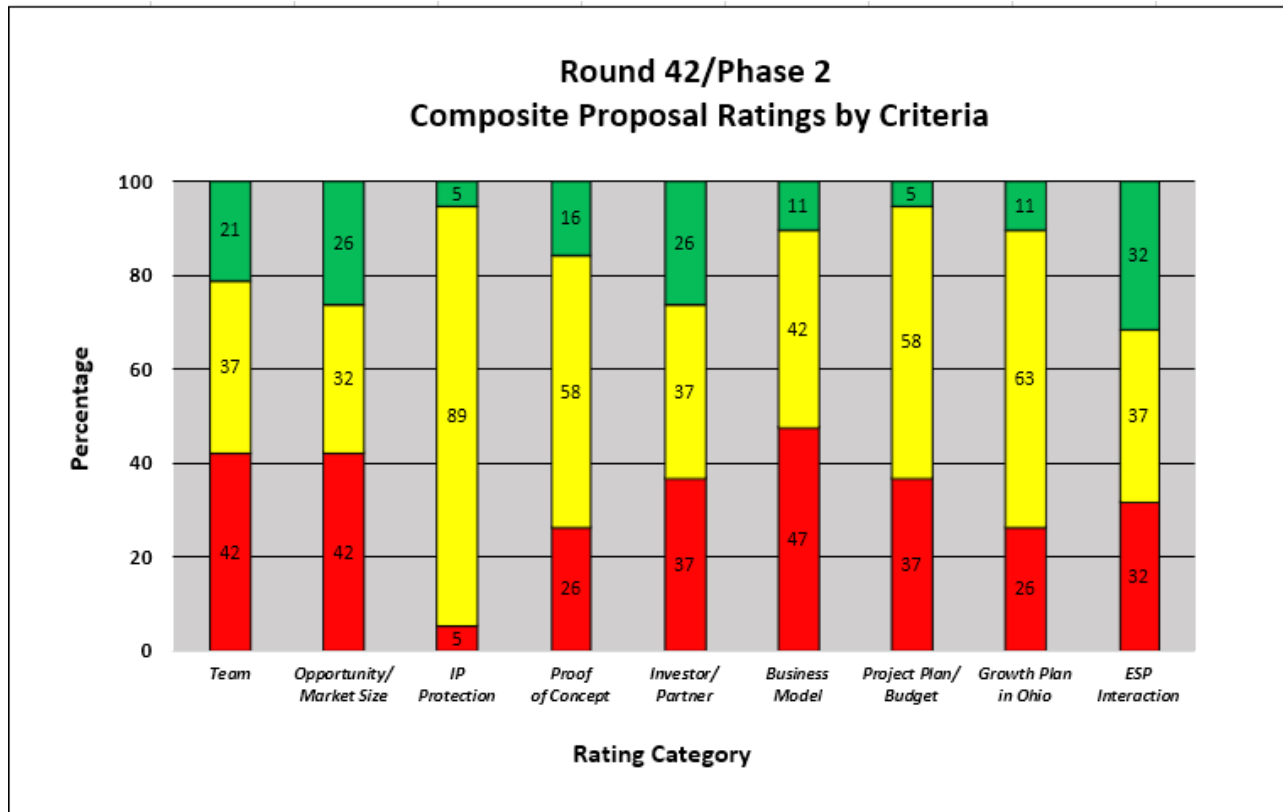
Figure 2. Round 42 Phase 2 Proposal Activity by Third Frontier Technology Area



TECHNOLOGY VALIDATION AND STARTUP FUND

Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. The Intellectual Property Protection had the lowest percentage of weak (red) scores in this round. Business Model followed by Team and Opportunity/Market Size had the highest percentage of weak scores in this round.

Figure 3. Round 42 Phase 2 Proposal Rating Summary



Of the previous twenty-two Rounds, business model was the lowest rating in Rounds 20-23 (53% average $\geq$ meets) and Round 26 (28%). The RFP was revised prior to Round 24 to elicit stronger business models and it appears that the proposals have provided stronger business models in Rounds 24-39 (68% average $\geq$ meets). Rounds 20 to 42 the overall average is 64%. Round 26 appears to be an anomaly. Future rounds will be monitored to see if the improvement continues.

Figure 4: Rounds 20 to 42 Phase 2 Analysis of Business Model

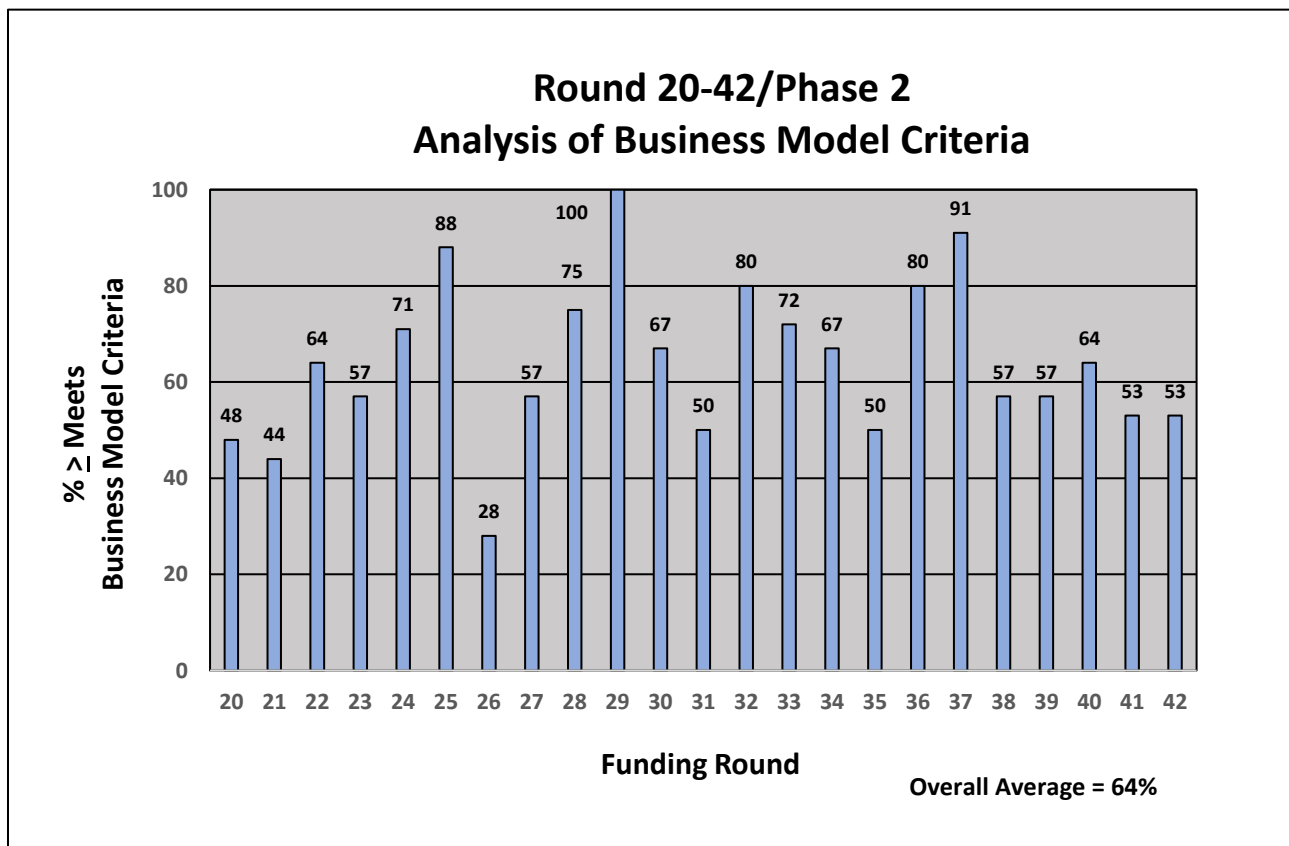
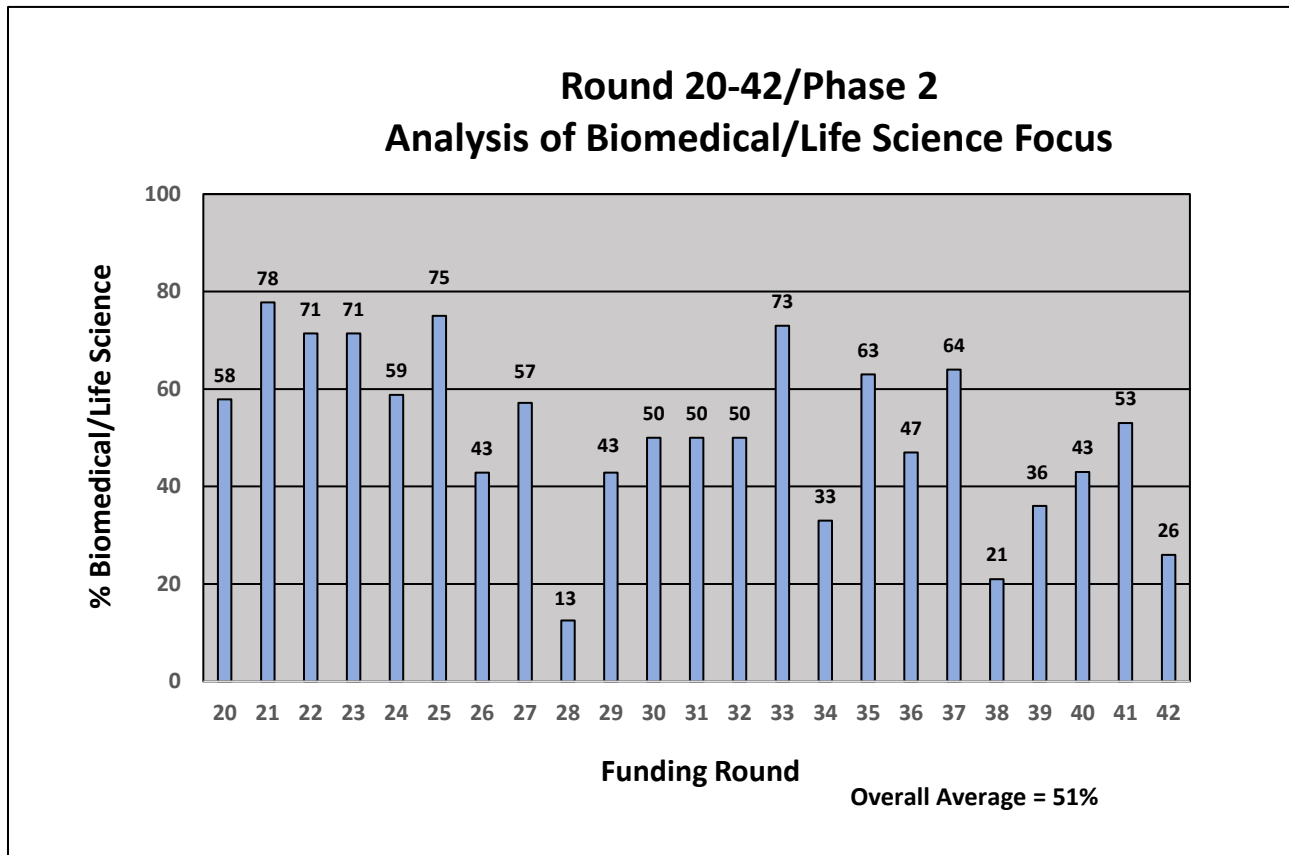


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last twenty-three Rounds. Biomedical/Life Sciences was the focus area for 26% of Round 42 applications. The overall average is 51% of the applications in Biomedical/Life Sciences.

Figure 5: Rounds 20-42 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There were 3 Phase 2 applicants that previously received Phase 1 funding. None of those applications are recommended for funding.

Phase 2: There were 4 Phase 2 reapplications in this Round. All four were first time reapplications and two are recommended for funding.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 12 years while each team member has been active in this field for over 25 years.

Each Redwood team member:

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The five members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

Market/Technology Assessment (140+ years)

Startup/ Spin out companies (50+ years)

Board member/Advisor to Startups (30+ years)

Evaluating/ monitoring RFPs/ Funding selection (40+ years)

The following is a brief summary of the five principal team members used in this evaluation Round.

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to \$52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

Jeff Crompton

- M.A., D.Phil., University of Oxford, Materials Science
- Founder and Former CEO AltaSim Technologies LLC
- Former Market and Technology Leader, EWI
- Former Technology Manager, Calgon Specialty Chemicals
- Former Principal Scientist, Alcan International
- Led development of innovative technology for computational analysis, medical products, aerospace and automotive markets
- Led commercial implementation of technology in production operations in North America, Europe and Asia
- Former Wolfson Industrial Fellow, University of Oxford

John McArdle

- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance / International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPh, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents
- Former VP Market and Technology Assessment at the National Technology Transfer Center
- Over 15 years as an angel investor in technology-based startups
- Over 15 years as an evaluator for Ohio Third Frontier funds including IOF, CALF and TIP
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/ proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent third-party source.

The goals of Phase 2 include:

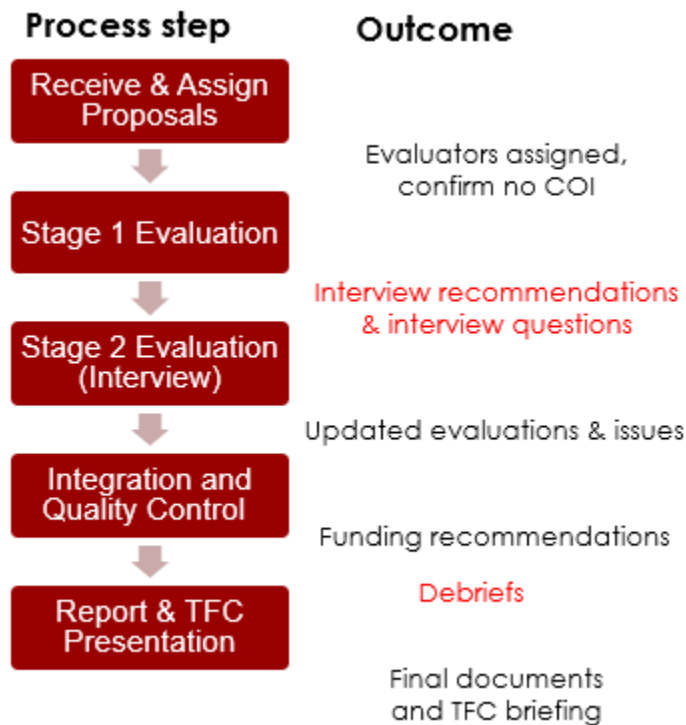
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation, and market research/ business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

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Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person, 45-minute interviews were held with each invited applicant to discuss the advance questions and other topics of interest to the evaluators. Two Redwood team members participated in each interview video conference, with additional team members joining, as needed.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go / No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3 rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

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Table 2 – Phase 2 Evaluation Criteria

Criterion	Weighting	Description
Alignment & compliance	Go / No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity / market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection / License	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition, is license likely within 9 months?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor / Business Partner Engagement	10	Is there company engagement / collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service / manufacturing model be scaled?
Project Plan / Budget Narrative	5	Is the budget consistent with proof in 1 year?
Start-up in Ohio	5	Does a start-up exist or is it planned? Will the start-up be in Ohio?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?